1. A magnetic core with a uniform cross-sectional area has a single winding. The core is driven into saturation. If the current in the winding is doubled, what happens to the flux density (B) and the inductance (L) of the winding?
   A) B doubles, L remains constant.
   B) B increases slightly, L decreases significantly.
   C) B doubles, L halves.
   D) B increases slightly, L remains constant.
   E) B doubles, L also doubles.
2. A simple rotating loop (the elementary DC machine) is operating as a motor. The magnetic field is uniform. At the instant the plane of the loop is perpendicular to the magnetic field lines ($\theta = 0°$), what is the value of the induced torque?
   A) Maximum
   B) Zero
   C) Half of its maximum value
   D) Equal to the load torque
   E) Cannot be determined
3. Two otherwise identical ferromagnetic cores are wound with the same number of turns. Core A has a large air gap, while Core B has a very small air gap. When the same current is applied to both windings, which core will have a higher stored magnetic field energy, and which will have a higher flux density?
   A) Core A has higher energy and higher flux density.
   B) Core B has higher energy and higher flux density.
   C) Core A has higher energy, but Core B has higher flux density.
   D) Core B has higher energy, but Core A has higher flux density.
   E) Both have the same energy and flux density.
4. Consider a linear DC machine acting as a motor. It is initially operating at a steady-state speed. If the polarity of the battery is suddenly reversed while the magnetic field direction remains the same, what is the immediate result?
   A) The motor continues running in the same direction at a slower speed.
   B) The motor instantly reverses direction.
   C) The machine acts as a generator, charging the battery.
   D) The bar decelerates to zero and then remains stationary.
   E) The bar decelerates to zero and then accelerates in the opposite direction.
5. In a DC machine, the hysteresis loss in the armature core is primarily dependent on which of the following factors?
   A) The thickness of the stator laminations.
   B) The square of the flux density and the frequency of magnetic reversal.
   C) The resistivity of the copper windings.
   D) The armature current and the terminal voltage.
   E) The commutator's surface roughness.
6. The primary reason for constructing the armature core of a DC machine from thin, insulated laminations of silicon steel is to:
   A) Decrease the hysteresis loss to improve heat dissipation.
   B) Maximize the flux density in the core.
   C) Reduce the eddy current loss by increasing the resistance of the core path.
   D) Decrease the cost of manufacturing the machine.
   E) Provide mechanical strength against centrifugal forces.
7. A DC machine exhibits significant armature reaction under heavy load. If the brushes are shifted forward (in the direction of rotation) to compensate for poor commutation, the most likely consequence is:
   A) Reduced demagnetizing component of armature reaction
   B) Increased flux per pole and higher induced emf
   C) Reduced back emf and higher armature current
   D) Improved commutation but reduced generated emf
   E) Increased torque with improved efficiency
8. In a DC machine, if armature reaction is fully neutralized by compensating windings and interpoles, but the machine still exhibits sparking at the brushes, the most probable cause is:
   A) High armature resistance

B) Saturation under the pole faces
C) Improper brush contact or segment roughness
D) Excessive residual magnetism
E) Low field current

9. The voltage regulation of a shunt DC generator is calculated as $VR = \frac{V_{nl} - V_{fl}}{V_{fl}}$ If a generator has significant armature reaction, its voltage regulation compared to the case with no armature reaction will be:
A) Better (a lower positive value or more negative value).
B) Worse (a higher positive value).
C) Unchanged.
D) Impossible to determine without the OCC.
E) Zero in both cases.

10. The process of "flashing the field" in a shunt DC generator that has failed to build up voltage is performed to:
A) Burn off contamination from the commutator.
B) Restore the residual magnetism in the pole faces.
C) Test the insulation strength of the field winding.
D) Align the magnetic domains in the armature core.
E) Calibrate the no-volt release coil.

11. The most significant factor limiting the maximum power output of a DC shunt generator is:
A) Armature heating due to I²R losses
B) Field resistance
C) Magnetic saturation
D) Friction and windage losses
E) Armature reaction

12. For a DC shunt motor (with constant field flux), a plot of developed torque vs. speed (T–ω) has a negative slope. This characteristic is inherently due to:
A) Field weakening with speed
B) The presence of the shunt field.
C) No change in flux
D) Armature reaction increasing linearly
E) The inverse relationship between speed and armature current caused by loading.

13. A 6-pole DC machine has a lap winding with 600 conductors (Z=600). The speed is 900 rpm, and flux per pole is 15 mWb. Find the generated emf.
A) 90 V
B) 120 V
C) 135 V
D) 150 V
E) 180 V

14. In a DC shunt motor, reducing the field current by 25% while keeping armature voltage constant will approximately:
A) Increase speed by 33%
B) Decrease torque by 25%
C) Increase speed by 25%
D) Increase torque by 33%
E) Keep speed constant

15. A DC series motor draws 40 A at 400 rpm while developing 100 Nm torque. Neglecting losses, if current increases to 60 A, the approximate speed is:
A) 150 rpm
B) 267 rpm
C) 290 rpm
D) 600 rpm
E) 900 rpm

16. A 250 V cumulative compound DC motor has the following data: resistances of armature, shunt field, and series field are 0.2 Ω, 250 Ω, and 0.1 Ω, respectively. Rated line current I_L is 50 A. Flux due to the shunt field alone corresponds to a speed of 1200 rpm at rated voltage (no load). The flux increases 10% when the motor operates at full load due to the cumulative effect of the series field. Neglect mechanical losses. Find the approximate full-load speed of the motor.
A) 1030 rpm
B) 1070 rpm
C) 1110 rpm
D) 1150 rpm
E) 1200 rpm

17. A 15-kW, 240 V, 1800 r/min separately excited DC generator has the following parameters and test data: Armature resistance (R_A) = 0.15 Ω, total brush voltage drop (V_{brush}) = 1.5 V. The generator's open-circuit characteristic (OCC) at 1800 r/min is given below:

Field current, I_F (A).	0.8.	1.2.	1.6.	2.0.	2.4
Internal voltage, E_A (V)	180	240	285	320	345

Under full-load conditions, the terminal voltage is measured to be $V_T = 230\ V$. The approximate full-load armature current I_A and the corresponding effective field current I_F, eff.

A) $I_A = 65.2\ A, I_{F,eff} = 1.00\ A$

B) $I_A = 65.2\ A, I_{F,eff} = 1.20\ A$

C) $I_A = 65.2\ A, I_{F,eff} = 1.40\ A$

D) $I_A = 62.5\ A, I_{F,eff} = 1.25\ A$

E) $I_A = 62.5\ A, I_{F,eff} = 1.60\ A$

18. A DC shunt motor operates under constant armature voltage. If the field resistance is increased by 20%, and load torque remains constant, which of the following statements is most accurate?

A) Armature current decreases and speed increases.

B) Armature current increases slightly and speed increases.

C) Both armature current and torque remain constant.

D) Flux increases causing speed to decrease.

E) The motor slows down to protect the commutator.

19. A DC shunt generator has a rated voltage of 250 V and a terminal voltage of 230 V at full load. If the voltage drop ($R_A I_A$) due to armature resistance is 5 V what portion of the voltage drop is caused by armature reaction?

A) 5 V

B) 10 V

C) 15

D) 20 V

E) Cannot be determined

20. A differentially compounded DC generator delivers power to a variable load. As the load current increases, the terminal voltage tends to:

A) Increase linearly with load

B) Remain constant

C) Drop sharply due to opposing series field mmf

D) Increase slightly due to residual magnetism

E) Rise due to improved commutation

FORMULAS PROVIDED

$P = E_A I_A$ P_{conv} $= e_{ind}.i$ $= F_{ind}.v$	$P = EI,$ $F = i(lxB),$ $e_{ind} =$ $(v\ x\ B).l$ $\tau = K\emptyset I_A$	$E_A = K\emptyset w,$ $K = \frac{ZP}{2\pi a},$ $a = mP$	$Pout = w\ \tau$ $\Phi \propto I_A$ (series motor) $T \propto I_a^2$ (series motor)

Power & Energy Conversion:

`P_elec = E_A * I_A`

`P_conv = P_mech = E_A * I_A = τ * ω`

Fundamentals:

`F = i * l * B` (Force on a conductor)

`e = B * l * v` (Induced EMF in a conductor)

Macroscopic Equations:

`E_A = K * φ * ω`

`τ = K * φ * I_A`

`K = (Z * P) / (2 * π * a)`

Key Relationships:

- **Series Motor:** `φ ∝ I_A`, `T ∝ I_A²`
- **Generator Voltage Drop:** `ΔV = I_A R_A + (Armature Reaction Drop)`